Chapter 1.1 Probability Spaces

Rick Durrett Probability: Theory and Examples

MSU Probability Prelim

Built from the local Chapter 1 LLMviki flash cards.

Roadmap for 1.1

Learning goal

Speak fluently about events, sigma-fields, probability measures, and extension from simple generating classes.

1. Probability spaces and sigma-fields.
2. Continuity of measure.
3. Discrete and Stieltjes probability models.
4. Extension from semialgebras and rectangles.
5. Exercise patterns: generators, increasing sigma-fields, density counterexamples.

Probability Space

Definition

A probability space is a triple
$$(\Omega, \mathcal{F}, \mathbb{P}),$$
where Ω is the sample space, $\mathcal{F}$ is a sigma-field of events, and $\mathbb{P}$ is a countably additive measure with $\mathbb{P}(\Omega) = 1$.

- Events are only the subsets in $\mathcal{F}$.
- Most elementary probability identities come from closure of $\mathcal{F}$ and countable additivity of $\mathbb{P}$.

LLMviki: `Probability space; ID: durrett_ch1_probability_space`

Sigma-Field Closure Rules

Definition

A class $\mathcal{F}$ is a sigma-field if:

$$\Omega \in \mathcal{F}, \qquad A \in \mathcal{F} \Rightarrow A^c \in \mathcal{F}, \qquad A_1, A_2, \ldots \in \mathcal{F} \Rightarrow \bigcup_{n=1}^{\infty} A_n \in \mathcal{F}.$$

Consequences

$\mathcal{F}$ is also closed under countable intersections, finite unions, finite intersections, and set differences.

LLMviki: `Sigma-field closure rules; ID: durrett_ch1_sigma_field`

How to Prove Sigma-Field Closure Consequences

Target

Derive countable intersection closure from complement and countable union closure.

LLMviki: `Sigma-field closure rules; ID: durrett_ch1_sigma_field`

Basic Probability Identities

For $A, B \in \mathcal{F}$:

$$\mathbb{P}(A^c) = 1 - \mathbb{P}(A), \qquad A \subseteq B \Rightarrow \mathbb{P}(A) \leq \mathbb{P}(B),$$
$$\mathbb{P}(B \setminus A) = \mathbb{P}(B) - \mathbb{P}(A) \quad (A \subseteq B),$$
$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B) - \mathbb{P}(A \cap B).$$

Teaching point

These are not separate axioms. They are quick consequences of disjoint decomposition and countable additivity.

LLMviki: `Probability space; ID: durrett_ch1_probability_space`

Continuity of Measure

Theorem

If $A_n \uparrow A$, meaning $A_1 \subseteq A_2 \subseteq \cdots$ and $A = \bigcup_n A_n$, then

$$\mathbb{P}(A_n) \uparrow \mathbb{P}(A).$$

If $A_n \downarrow A$, meaning $A_1 \supseteq A_2 \supseteq \cdots$ and $A = \bigcap_n A_n$, then

$$\mathbb{P}(A_n) \downarrow \mathbb{P}(A).$$

- For general measures, the decreasing version needs $\mu(A_1) < \infty$.
- Under probability measures, this finiteness is automatic.

LLMviki: `Continuity of measure; ID: durrett_ch1_measure_continuity`

Proof Idea: Continuity from Below

Key move

Turn nested sets into disjoint increments.

LLMviki: `Continuity of measure; ID: durrett_ch1_measure_continuity`

Proof Idea: Continuity from Above

Key move

Reduce decreasing sets to increasing complements.

LLMviki: `Continuity of measure; ID: durrett_ch1_measure_continuity`

Discrete Probability Spaces

Model

If Ω is countable and $p(\omega) \geq 0$ with

$$\sum_{\omega \in \Omega} p(\omega) = 1,$$

then define

$$\mathbb{P}(A) = \sum_{\omega \in A} p(\omega), \qquad A \subseteq \Omega.$$

- Countable additivity follows from rearranging nonnegative series.
- This intuition does not determine laws on an uncountable space.

LLMviki: `Discrete probability spaces; ID: durrett_ch1_discrete_probability_space`

Stieltjes Measures from Distribution-Like Functions

Theorem

A nondecreasing right-continuous function $F : \mathbb{R} \to \mathbb{R}$ with suitable limits defines a unique Borel measure through interval increments:

$$\mu((a, b]) = F(b) - F(a).$$

If $F(-\infty) = 0$ and $F(\infty) = 1$, then μ is a probability measure.

Why it matters

This is the bridge from CDFs to probability measures on $\mathbb{R}$.

LLMviki: `Stieltjes measures from distribution-like functions; ID: durrett_ch1_stieltjes_measure`

How to Prove a Stieltjes Construction

1. Define $\mu((a,b]) = F(b) - F(a)$ on half-open intervals.
2. Verify nonnegativity and finite additivity on disjoint interval decompositions.
3. Use right-continuity of F to get the needed continuity property.
4. Extend from the interval semialgebra to $\mathcal{B}(\mathbb{R})$.
5. Use uniqueness because intervals generate the Borel sigma-field.

LLMviki: `Stieltjes measures from distribution-like functions; ID: durrett_ch1_stieltjes_measure`

Extension from Semialgebras

Proof template

A measure-like function on a semialgebra can be extended when it has the right countable additivity or continuity behavior on the generating class.

1. Start with simple pieces in $\mathcal{S}$.
2. Pass to finite disjoint unions of pieces.
3. Check countable additivity or continuity.
4. Extend to $\sigma(\mathcal{S})$.

LLMviki: `Extension from semialgebras; ID: durrett_ch1_semialgebra_extension`

Multidimensional Distribution Functions

Theorem

A function F on $\mathbb{R}^d$ satisfying right-continuity, correct limits, and nonnegative rectangle increments defines a probability measure on $\mathcal{B}(\mathbb{R}^d)$.

$$\mu\big((a_1, b_1] \times \cdots \times (a_d, b_d]\big) = \Delta_{a,b} F \geq 0.$$

Pitfall

Coordinatewise monotonicity is not enough. Rectangle probabilities must be nonnegative.

LLMviki: `Multidimensional distribution functions; ID: durrett_ch1_product_distribution_function`

How to Prove Generated Sigma-Field Claims

Standard two-inclusion proof

To show $\sigma(\mathcal{C}) = \mathcal{B}(\mathbb{R}^d)$:

1. Show every set in $\mathcal{C}$ is Borel, so $\sigma(\mathcal{C}) \subseteq \mathcal{B}(\mathbb{R}^d)$.
2. Show every open set belongs to $\sigma(\mathcal{C})$.
3. Use a countable rational base: every open set is a countable union of rational rectangles or balls.

LLMviki: `Sigma-field closure rules; ID: durrett_ch1_sigma_field`

LLMviki: `Extension from semialgebras; ID: durrett_ch1_semialgebra_extension`

Exercise 1: Countable/Co-countable Probability Space

Let $\Omega = \mathbb{R}$, let $\mathcal{F}$ be the collection of all subsets A such that A or A^c is countable, and define

$$\mathbb{P}(A) = \begin{cases} 0, & A \text{ is countable,} \\ 1, & A^c \text{ is countable.} \end{cases}$$

Show that $(\Omega, \mathcal{F}, \mathbb{P})$ is a probability space.

LLMviki: `Countable/co-countable probability measure; ID: exercise_method_countable_cocountable_measure`

Exercise 1: Proof Skeleton

LLMviki: Countable/co-countable probability measure; ID: exercise_method_countable_cocountable_measure

Exercise 2: Borel Field from Half-Open Rectangles

Recall the definition of $\mathcal{S}_d$ from Example 1.1.5:

$$\mathcal{S}_d = \{\emptyset\} \cup \{(a_1, b_1] \times \cdots \times (a_d, b_d] : -\infty \leq a_i < b_i \leq \infty\}.$$

Show that $\sigma(\mathcal{S}_d) = \mathcal{B}(\mathbb{R}^d)$, the Borel subsets of $\mathbb{R}^d$.

LLMviki: `Borel sigma-field from half-open rectangles; ID: exercise_method_borel_generated_by_half_open_rectangles`

Exercise 2: Proof Skeleton

LLMviki: Borel sigma-field from half-open rectangles; ID:
exercise_method_borel_generated_by_half_open_rectangles

Exercise 3: Countably Generated Borel Field

A sigma-field $\mathcal{F}$ is said to be countably generated if there is a countable collection $\mathcal{C} \subseteq \mathcal{F}$ such that $\sigma(\mathcal{C}) = \mathcal{F}$. Show that $\mathcal{B}(\mathbb{R}^d)$ is countably generated.

LLMviki: `Borel sigma-field from half-open rectangles; ID: exercise_method_borel_generated_by_half_open_rectangles`

Exercise 3: Proof Skeleton

LLMviki: `Borel sigma-field from half-open rectangles; ID: exercise_method_borel_generated_by_half_open_rectangles`

Exercise 4: Increasing Union of Sigma-Fields

If $\mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots$ are sigma-algebras, then:

(i) Show that $\bigcup_i \mathcal{F}_i$ is an algebra.

(ii) Give an example to show that $\bigcup_i \mathcal{F}_i$ need not be a sigma-algebra.

LLMviki: `Increasing union of sigma-fields can fail countable closure; ID: exercise_method_increasing_sigma_fields_not_sigma_field`

Exercise 4: Counterexample

LLMviki: `Increasing union of sigma-fields can fail countable closure; ID: exercise_method_increasing_sigma_fields_not_sigma_field`

Exercise 5: Asymptotic Density Is Not an Algebra

A set $A \subseteq \{1, 2, \ldots\}$ is said to have asymptotic density θ if

$$\lim_{n \to \infty} \frac{|A \cap \{1, 2, \ldots, n\}|}{n} = \theta.$$

Let $\mathcal{A}$ be the collection of sets for which the asymptotic density exists. Is $\mathcal{A}$ a sigma-algebra? Is $\mathcal{A}$ an algebra?

LLMviki: `Block construction for asymptotic density counterexamples; ID: exercise_method_asymptotic_density_block_counterexample`

Exercise 5: Block Counterexample

LLMviki: Block construction for asymptotic density counterexamples; ID: exercise_method_asymptotic_density_block_counterexample

Checklist: How to Attack 1.1 Problems

1. Need to prove "probability space"? Check sigma-field, normalization, countable additivity.
2. Need to prove "generated sigma-field"? Use two inclusions and a countable rational base.
3. Need a limit of probabilities? Try continuity from below/above.
4. Need a counterexample to sigma-field closure? Use finite initial-coordinate sigma-fields.
5. Need a density-limit counterexample? Use rapidly growing blocks.

LLMviki: `Chapter 1.1 exercise method cards`

Mini Practice

1. If $A_n \uparrow A$, prove $\mathbb{P}(A \setminus A_n) \to 0$.
2. Show $\sigma((a, b] : a < b) = \mathcal{B}(\mathbb{R})$.
3. Construct a finite algebra on $\Omega = \{1, 2, 3, 4\}$ generated by $\{1, 2\}$.
4. Suppose $A_n \downarrow \varnothing$. Explain why $\mathbb{P}(A_n) \to 0$.
5. Explain why an uncountable union of measurable sets need not be covered by the sigma-field axioms.